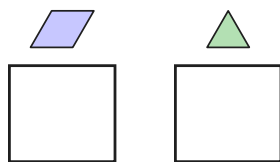
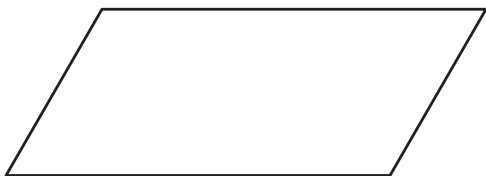
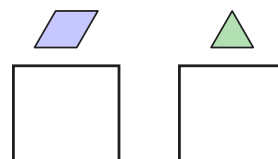
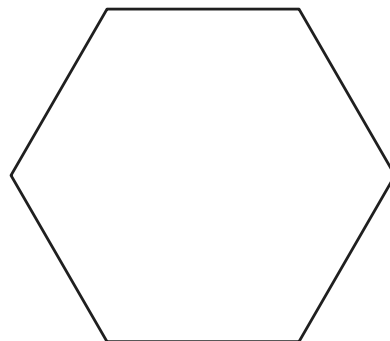
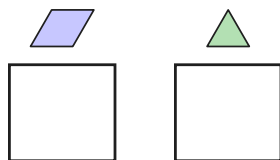
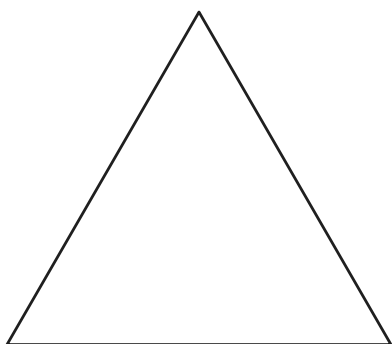
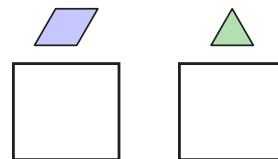
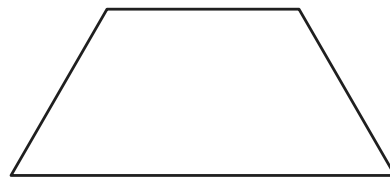
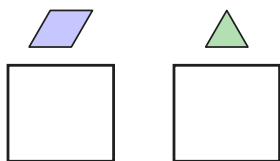
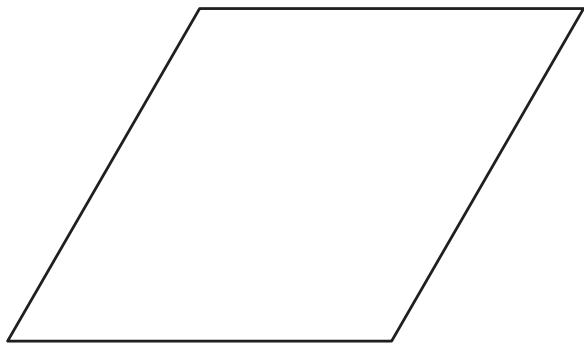
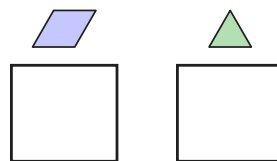
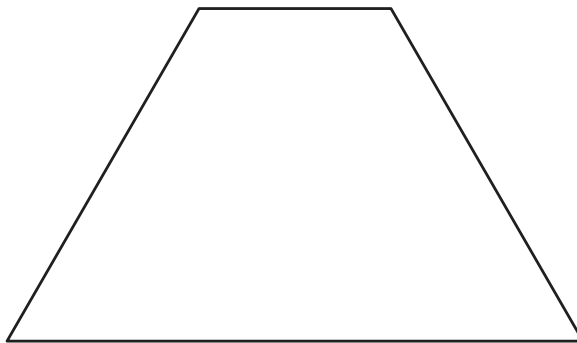
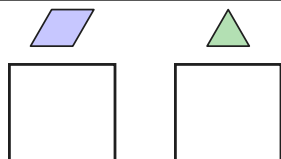
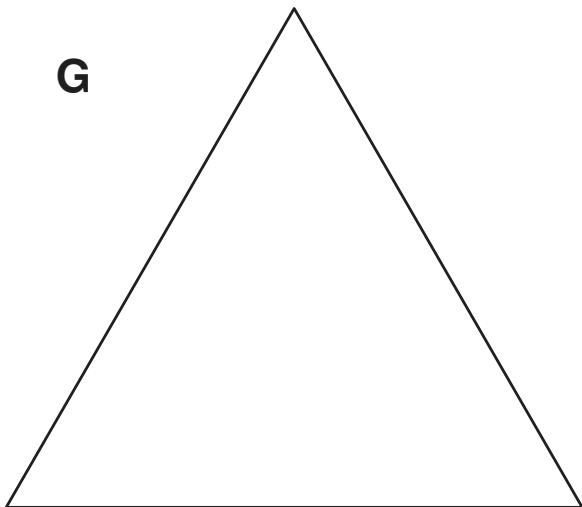
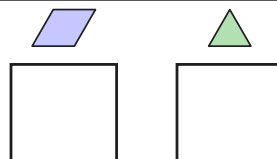
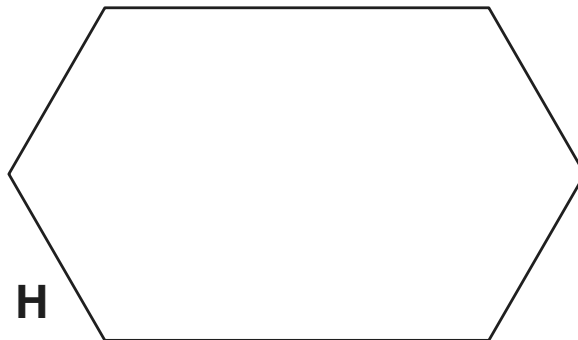


Problem 1: Cover each board with blue rhombi  only. Pieces must not overlap, leave gaps, or stick out past the outline. If you can cover the board, write $\checkmark$ in its first box; if you cannot, write $\times$. Then cover the same board with green triangles  and write how many you used in its second box.

A**B****C****D**

Problem 2: Do the same for boards E, F, G and H: blue rhombi  only, then $\checkmark$ or $\times$ in the first box; then green triangles , and the number in the second box.

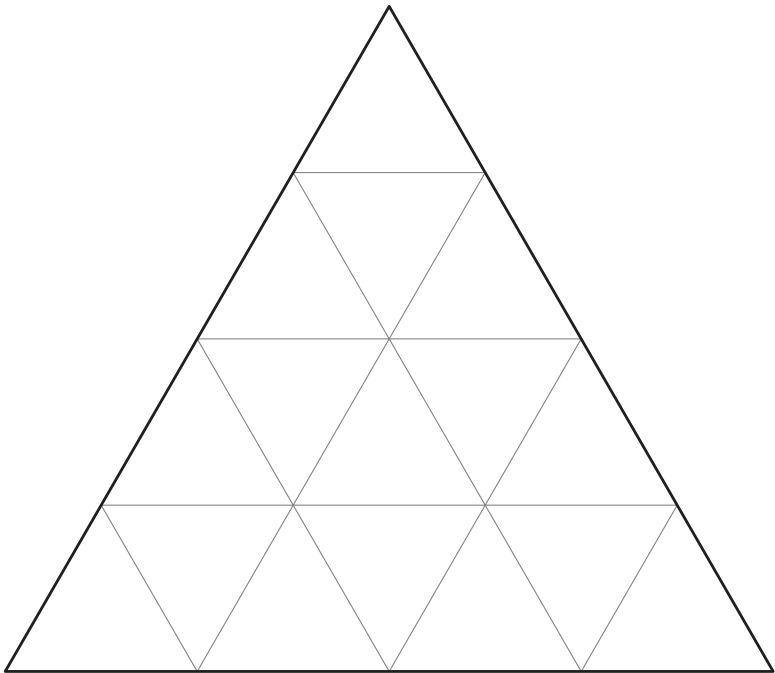
E**F****G****H**

Problem 3: Copy your results into the table: for each board, the number of green triangles and $\checkmark$ or $\times$. Circle every board whose number is odd. Then find two boards with the same number where one is $\checkmark$ and the other is $\times$, and write their letters in the two boxes.

board	green triangles	$\checkmark$ or $\times$
A		
B		
C		
D		
E		
F		
G		
H		

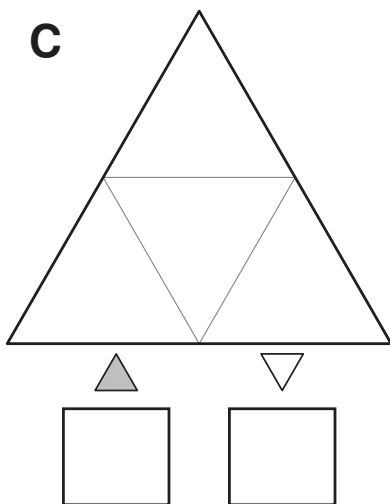
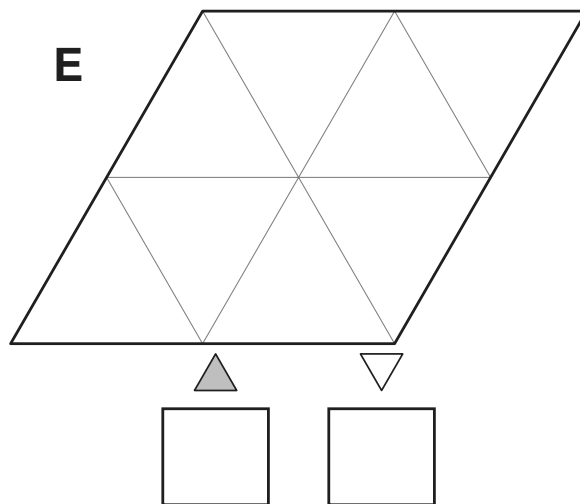
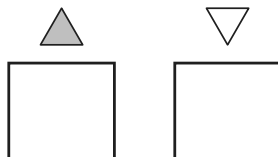
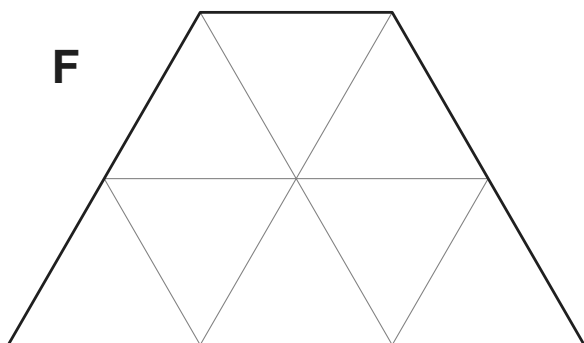
boards and

Problem 4: On the grid below, color every up-pointing triangle $\triangle$ and leave every down-pointing triangle ∇ white. A blue rhombus can lie on the grid in three directions. Put it on the grid in each direction so that it covers whole small triangles exactly, and each time write in the table how many colored and how many white triangles it covers. Then do the same with a red trapezoid, first with its long side at the bottom, then with its long side at the top; with a green triangle, pointing up and then pointing down; and with a yellow hexagon.



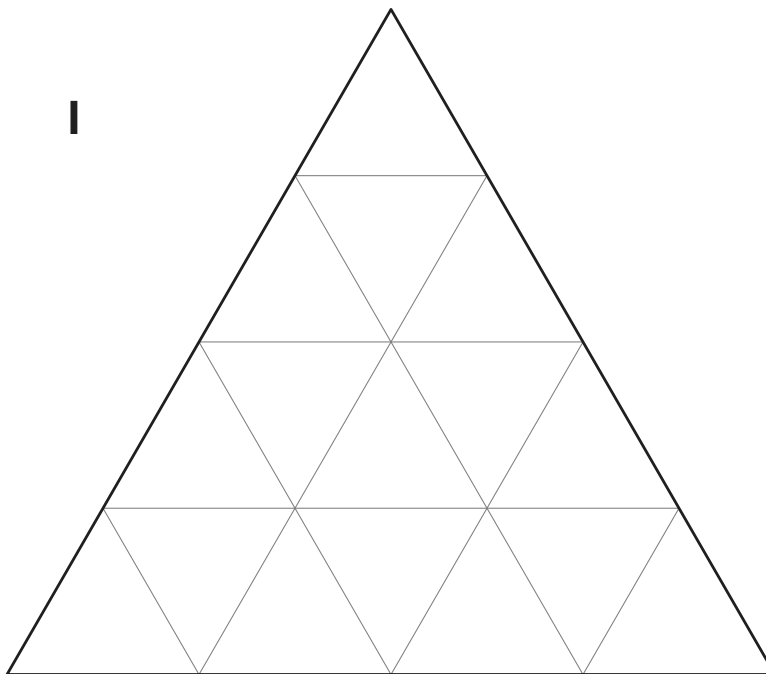
piece		
 direction 1		
 direction 2		
 direction 3		
 long side down		
 long side up		
 pointing up		
 pointing down		
		

Problem 5: Color the up-pointing triangles $\triangle$ in boards C, E and F. Count the colored and the white triangles in each board and write the two numbers in its boxes. Then explain on the lines why board F cannot be covered by blue rhombi.

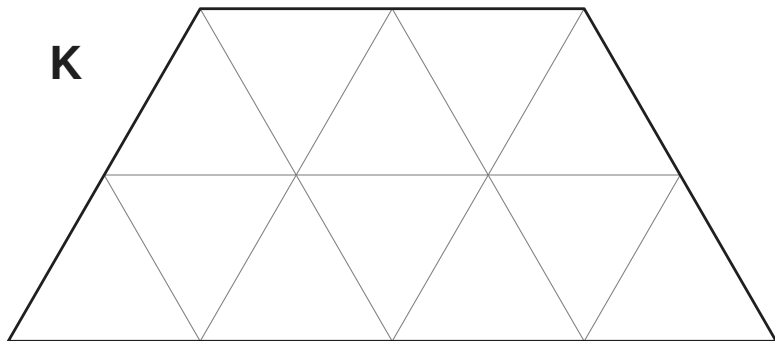
C**E****F**

Problem 6: For boards I and K: first color the up-pointing triangles and write the two counts in the table. Then write your prediction, $\checkmark$ or $\times$, before touching any pieces. Then check with blue rhombi and write $\checkmark$ or $\times$ in the result column.

I

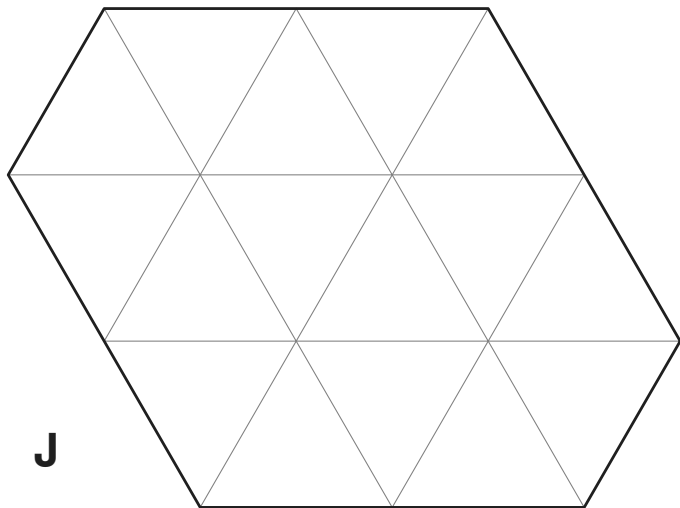


K

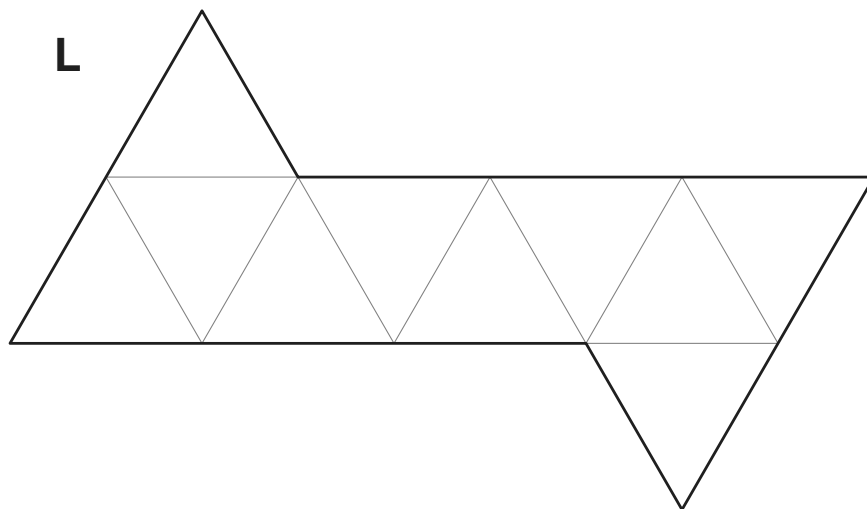


board			prediction	result
I				
K				

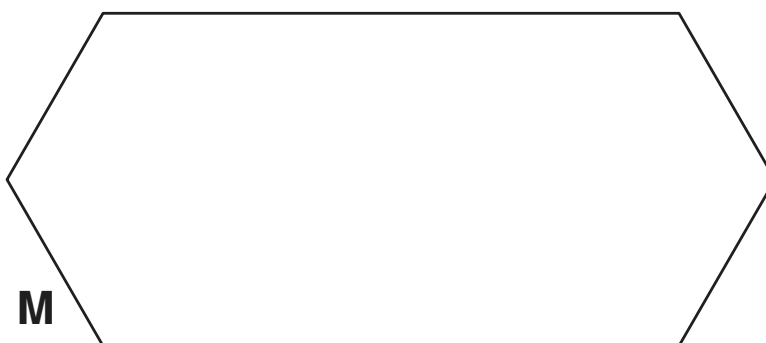
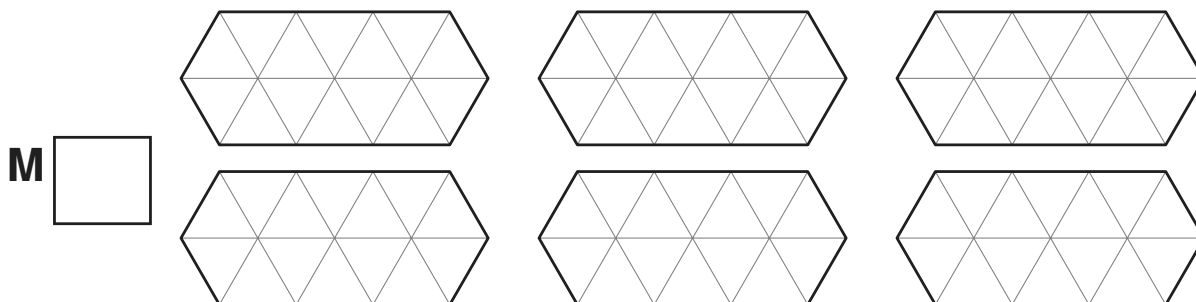
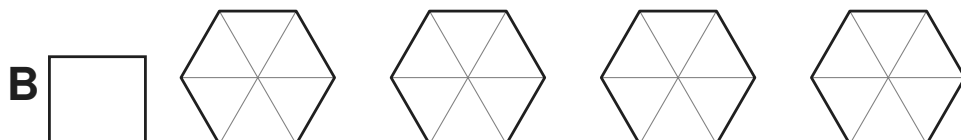
Problem 7: Do the same for boards J and L: color the up-pointing triangles, count, write your prediction, then check with blue rhombi and write $\checkmark$ or $\times$ in the result column. If your prediction was wrong for a board, write on the lines what you found when you tried to cover it.



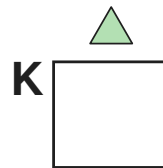
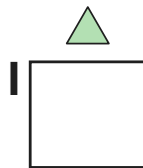
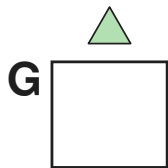
board			prediction	result
J				
L				



Problem 8: Board B can be covered by three blue rhombi in more than one way. Find every different way. Draw each way on one of the small copies of B by outlining each rhombus with a dark line, and write the number of ways in the box. Then do the same for board H (five rhombi) and for board M (seven rhombi). Board M is printed below at full size. There are more small copies than you need.



Problem 9: Cover board G (page 2) using blue rhombi  and green triangles  together, with as few green triangles as you can. Write your smallest number in the box for G. Do the same for board I (page 5) and board K (page 5). Then explain on the lines why nobody can cover board G with fewer green triangles than your number.



Problem 10: Which of the boards A to K can be covered with red trapezoids  only? Try each board and write ✓ or × in the box under its letter. Then write on the line one thing that is true about the number of green triangles of every board you could cover.

A	B	C	D	E	F	G	H	I	J	K
										
